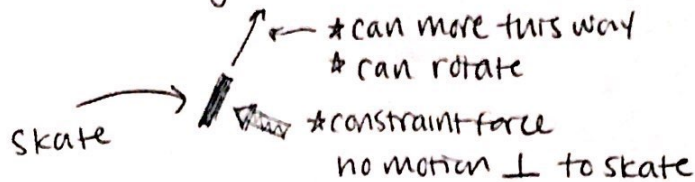


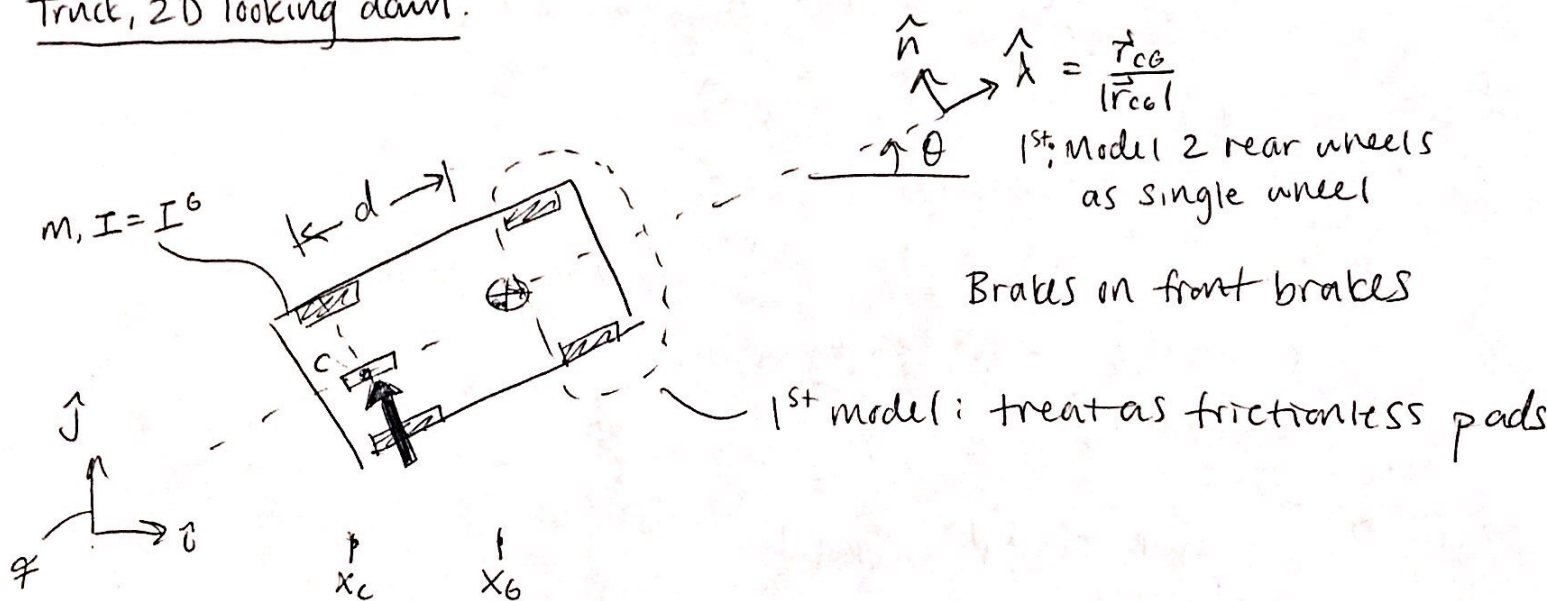
2D, looking down



Note:

- ideal skate = ideal (frictionless, neg. mass) wheel
- = wing = sail = keel
- = arrow feathers
- = butter knife in soft butter

Truck, 2D looking down:



Equations of Motion:

• Show us how to do problem:

• Want eq. w/o unknown force on rear wheel

$$\{A, M, B, c\} \cdot \hat{e} \quad (1)$$

$$\{L, M, B\} \quad (2)$$

• Variables to work with

$$AMB/c =$$

$$\sum \vec{M}_{/c} = \vec{H}_{/c}$$

$$\vec{0} = \vec{r}_{b/c} \times m \vec{a}_b + I \dot{\omega} \hat{k} \quad (1)$$

$$\{LMB\} \cdot \hat{\lambda} :$$

$$\{\sum \vec{F}_{ext} = \vec{L}\} \cdot \hat{\lambda}$$

$$0 = m \vec{a}_b \cdot \hat{\lambda} \quad (2)$$

$$\vec{r}_{b/c} = d \hat{\lambda}$$

$$\hookrightarrow \cos \theta \hat{i} + \sin \theta \hat{j} \quad (\text{turns out but don't need to know})$$

$$\omega = \dot{\theta} \quad (\text{also turns out don't need to know})$$

$$\vec{a}_b : (\text{tricky part of problem})$$

$$= \vec{a}_c + \vec{a}_{b/c}$$

$$\hookrightarrow \vec{a}_{b/c} = -\omega^2 d \hat{\lambda} + \dot{\omega} d \hat{n}$$

$$\vec{a}_c = \frac{d}{dt} (\vec{v}_c) = \frac{d}{dt} v_c \hat{\lambda} = \dot{v}_c \hat{\lambda} + v_c \dot{\hat{\lambda}} \quad \hookrightarrow \omega \hat{n}$$

$$\Rightarrow (1) \Rightarrow 0 = I \dot{\omega} \hat{k} + m \dot{\omega} d^2 + v_c \omega m d$$

$$\dot{\omega} = \left[\frac{-md}{I + md^2} \right] v_c \omega \quad c < 0 \text{ if } d > 0, \quad c > 0 \text{ if } d < 0$$

$$(2) \Rightarrow 0 = -\omega^2 d + \dot{v}_c \quad \Rightarrow \boxed{\dot{v}_c = +\omega^2 d}$$

$\Rightarrow$ dynamics

To find motion on plane

⇒ add more kinematic equations

$$\dot{\theta} = \omega$$

$$\begin{bmatrix} \dot{x}_c = v_c \cos \theta \\ \dot{y}_c = v_c \sin \theta \end{bmatrix} \Leftrightarrow \vec{v}_c = v_c \hat{\lambda}$$

look @ behavior of system in neighborhood of solution

Stability Analysis:

Find a nominal motion that satisfies equations

$$v_c = v_c^* = \text{constant}$$

$$\omega = 0$$

solves ① + ②

small perturbation, not unit vector

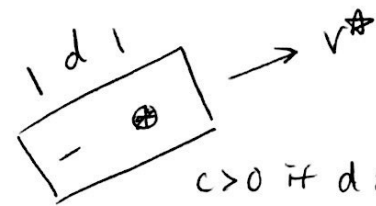
$$v_c = v_c^* + \hat{v}_c$$

↑
constant

$$\hat{v}_c \ll 1$$

plug into governing eqns

$$\omega = 0 + \hat{\omega}$$



$c > 0$ if d & v^* are opposite

$c < 0$ if d & v^* are same direction

$$\textcircled{2} \quad \dot{\hat{v}}_c = \underbrace{\hat{\omega}^2}_{\text{h.o.t.}} d \Rightarrow \hat{\omega}^2 = 0$$

$$\frac{d(0 + \hat{\omega})}{dt} = c(v_c^* + \hat{v}_c)(0 + \hat{\omega})$$

$$\dot{\hat{\omega}} = c v_c^* \hat{\omega} + \underbrace{c \hat{v}_c \hat{\omega}}_{\text{higher order term (h.o.t.)}} \\ \ll \hat{\omega} v^* \\ \Rightarrow \text{neglect}$$

⇒ make animations

Non linear

• interested in v_c, ω

• products / non linear

→ not like eg in system dynamics

• turns out simple to solve but not always the case

EOM linearized

$$\begin{aligned}\dot{\hat{\omega}} &= c V^* \hat{\omega} \\ \dot{\hat{V}}_c &= 0\end{aligned}$$

Linear
ODEs

$$\hat{\omega} = \hat{\omega}_i e^{c V^* t}$$

- exponential growth
- $c < 0 \Rightarrow$ decay